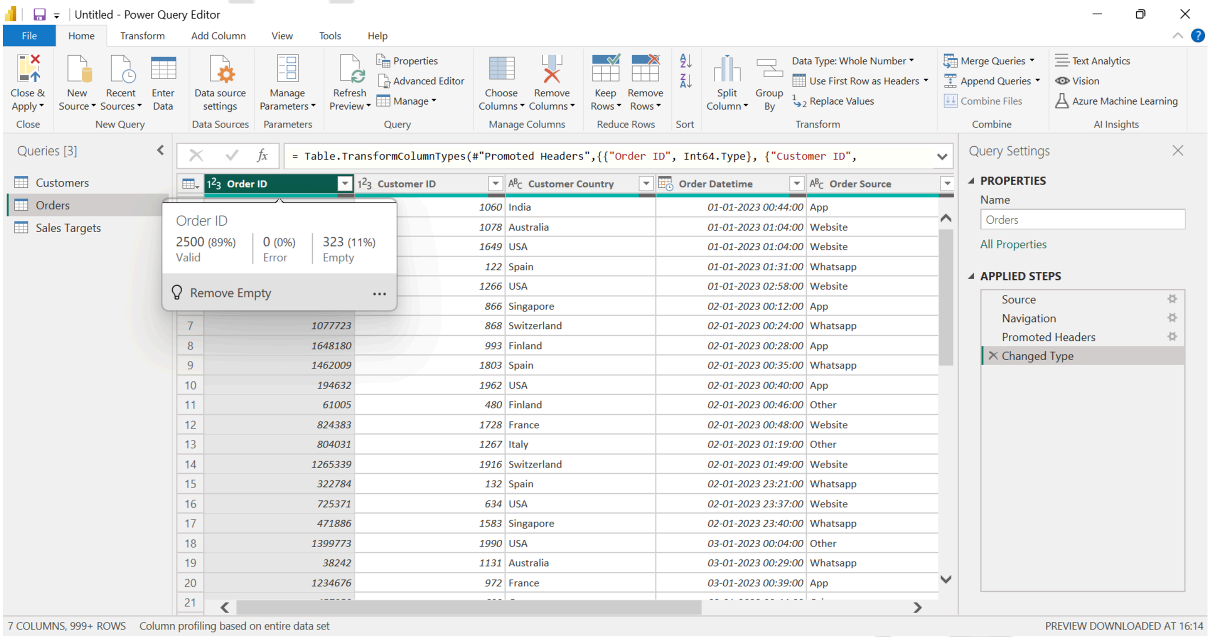
DIY PROJECT: E-COMMERCE ORDERS | SOLUTION

This document consists the stepwise solution for the Power BI Project and file below is the final PBIX Solution for the same:

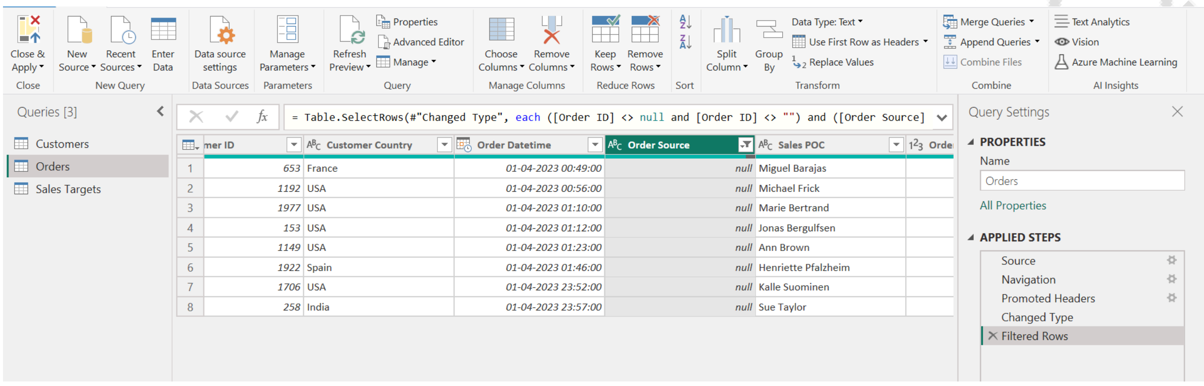
Link to Solution File(Open in a new Tab)

Solutions

1. After importing data, load Power Query Editor
   a. See if Power BI added extra nulls rows at the bottom of the dataset. If yes, remove them by using Remove Empty.

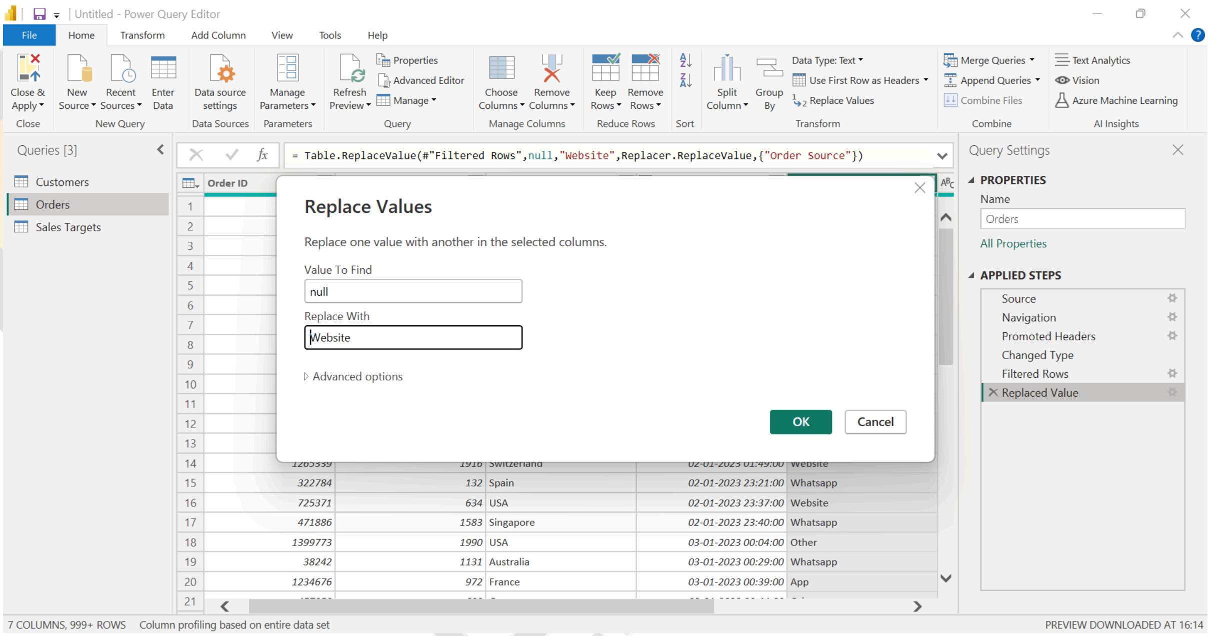


2. To see the missing order source (sales channel), use filter - filter order source for null values

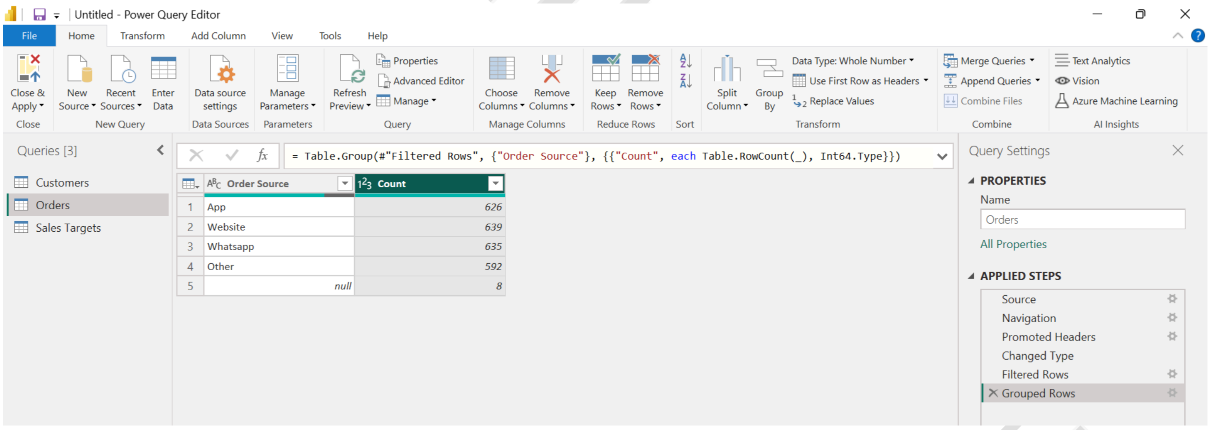


a. The first and last timestamps when we weren't capturing this data can be seen from the image above.

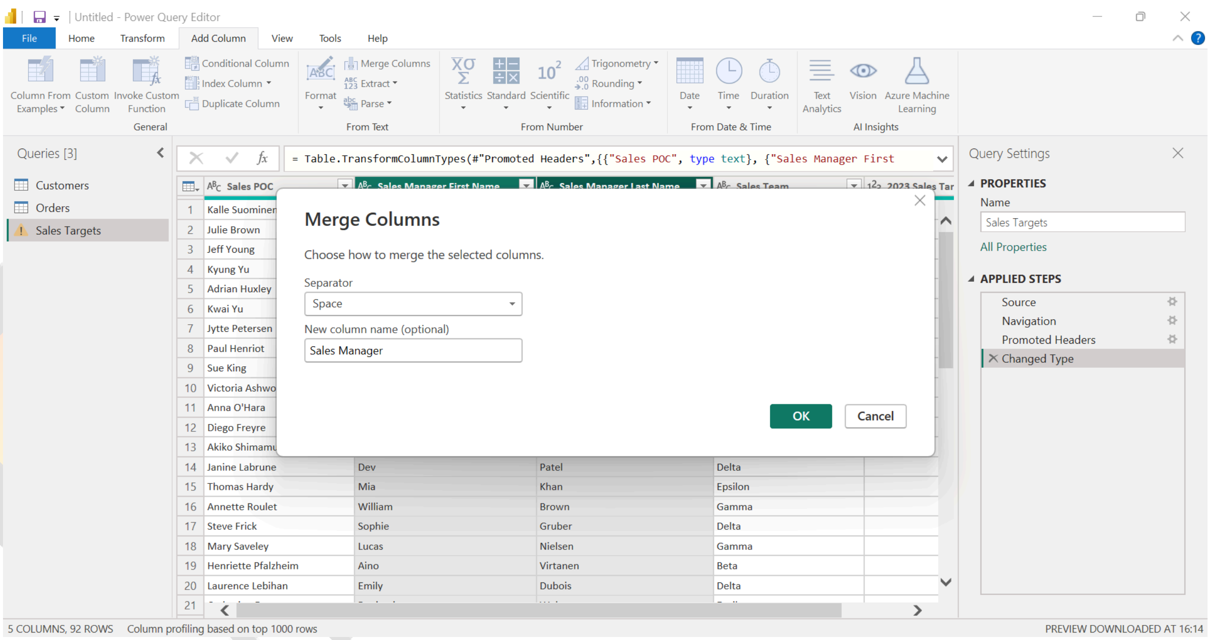
b. Replace these null values by the Order Source that occurs most frequently: i.e. Website.



c. **Note**: to find most occurring order source, we can use group by as shown here:

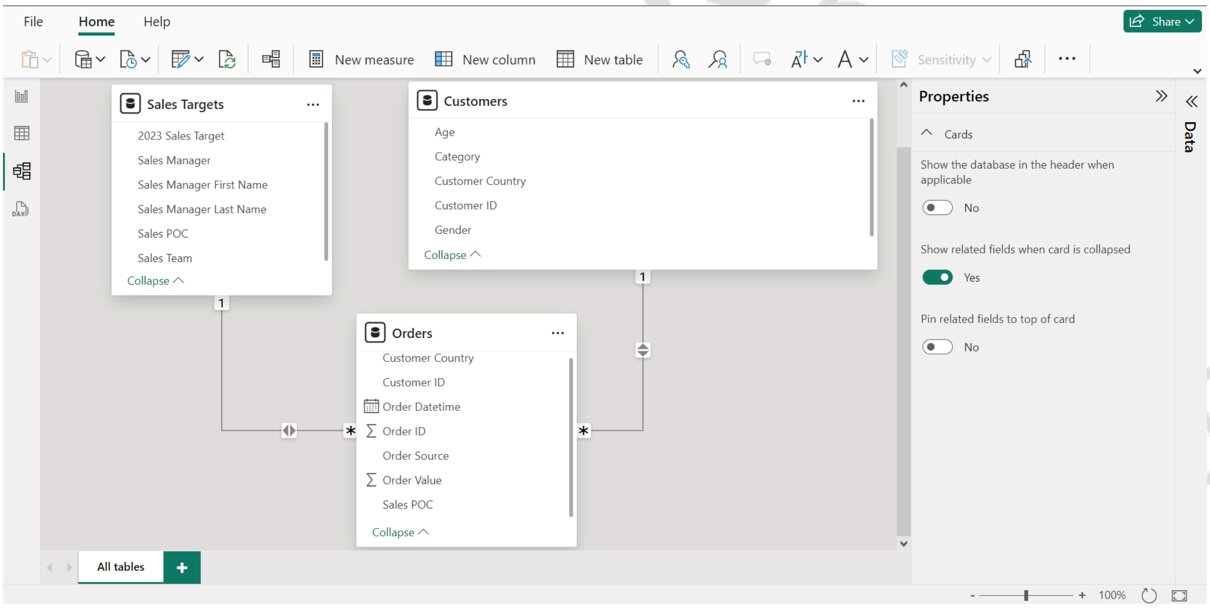


3. Use Add Column → Merge Columns to create a Sales Manager Column



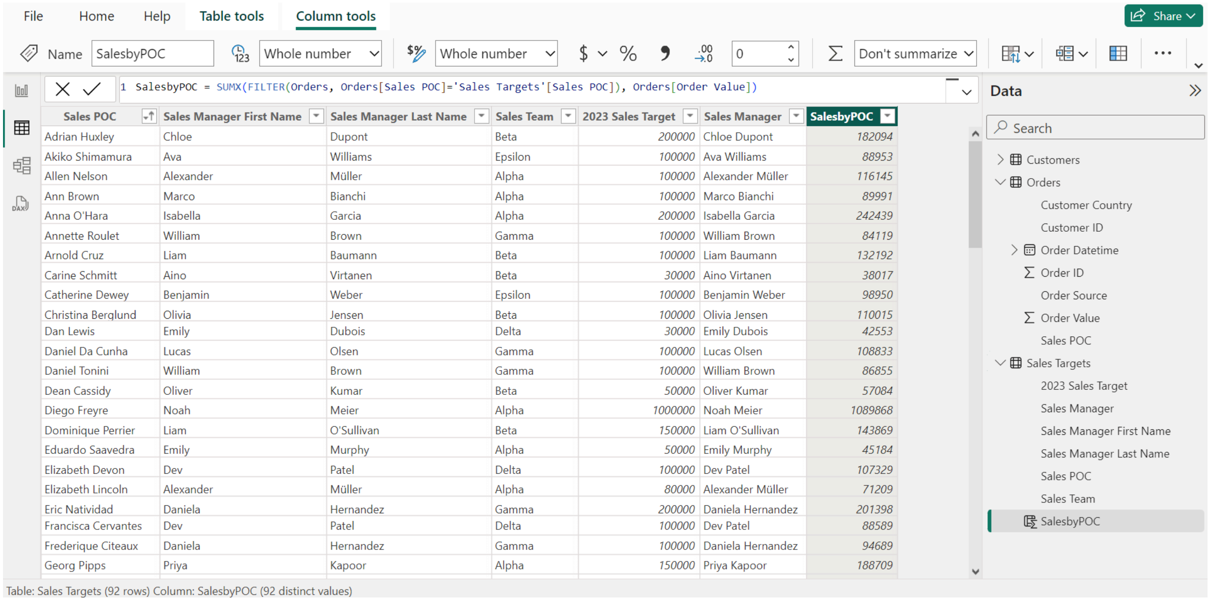
Load the data by clicking Close and Apply after this step.

4. Go to Model View and create relationships to ensure the following:
   a. Orders table is related to Sales Targets by Sales POC in a 1:many bidirectional relationship (from Sales Targets to Orders)
   b. Orders table is related to Customers by Sales POC in a 1:many single bidirectional relationship (from customers to Orders)
   c. Bidirectional is needed here because we want the filters to work in both directions

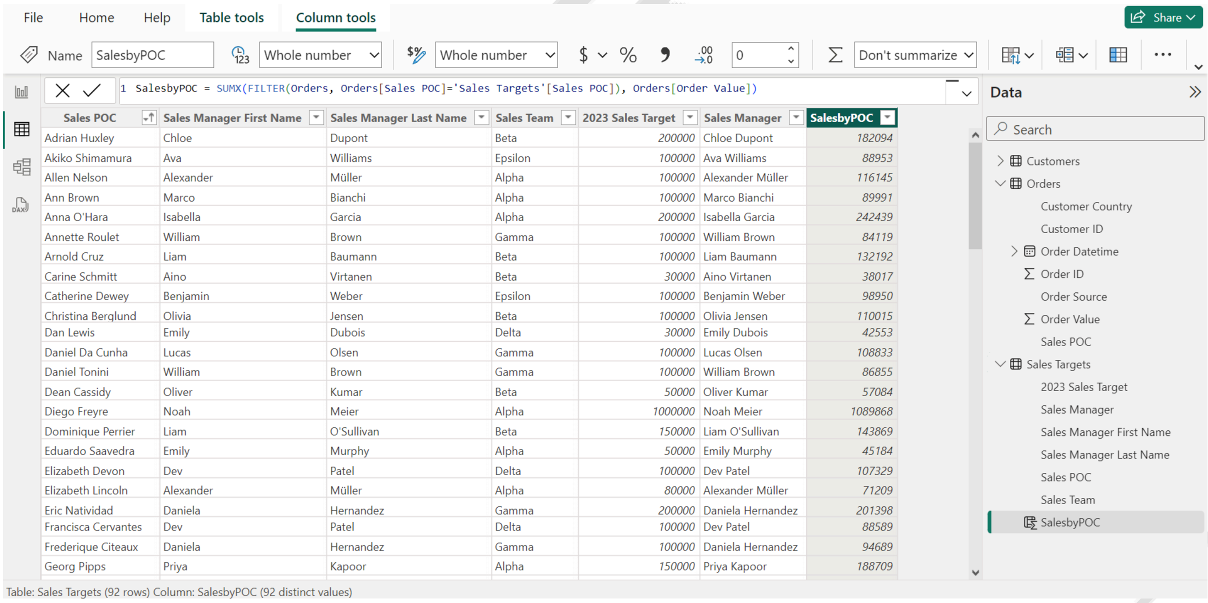


5.

a. Create a new column in the Sales Targets table that calculates Sales for each Sales POC



SalesbyPOC = SUMX(FILTER(Orders, Orders[Sales POC]='Sales Targets'[Sales POC]), Orders[Order Value])



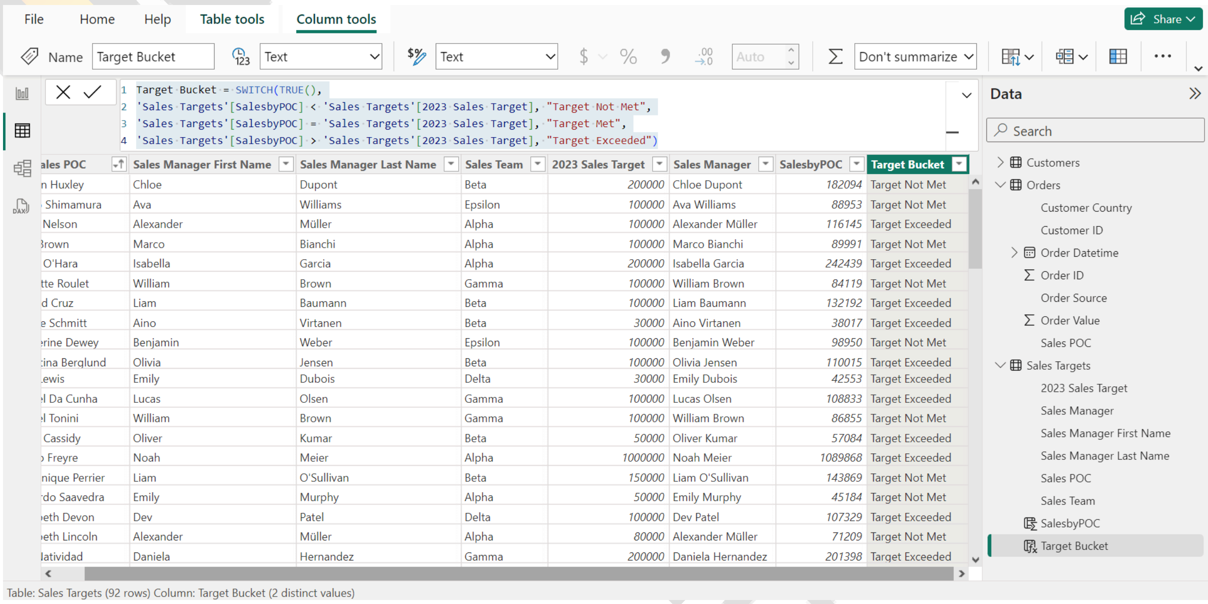
Use DAX SWITCH function to define three buckets.
This formula can be used:

Target Bucket = SWITCH(TRUE(),

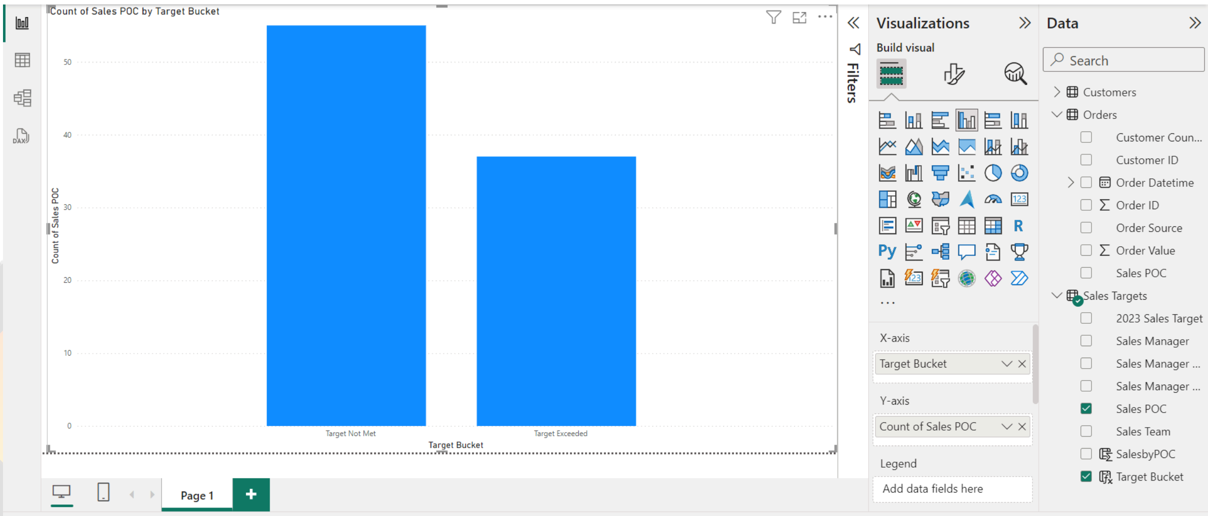
'Sales Targets'[SalesbyPOC] < 'Sales Targets'[2023 Sales Target], "Target Not Met",

'Sales Targets'[SalesbyPOC] = 'Sales Targets'[2023 Sales Target], "Target Met",

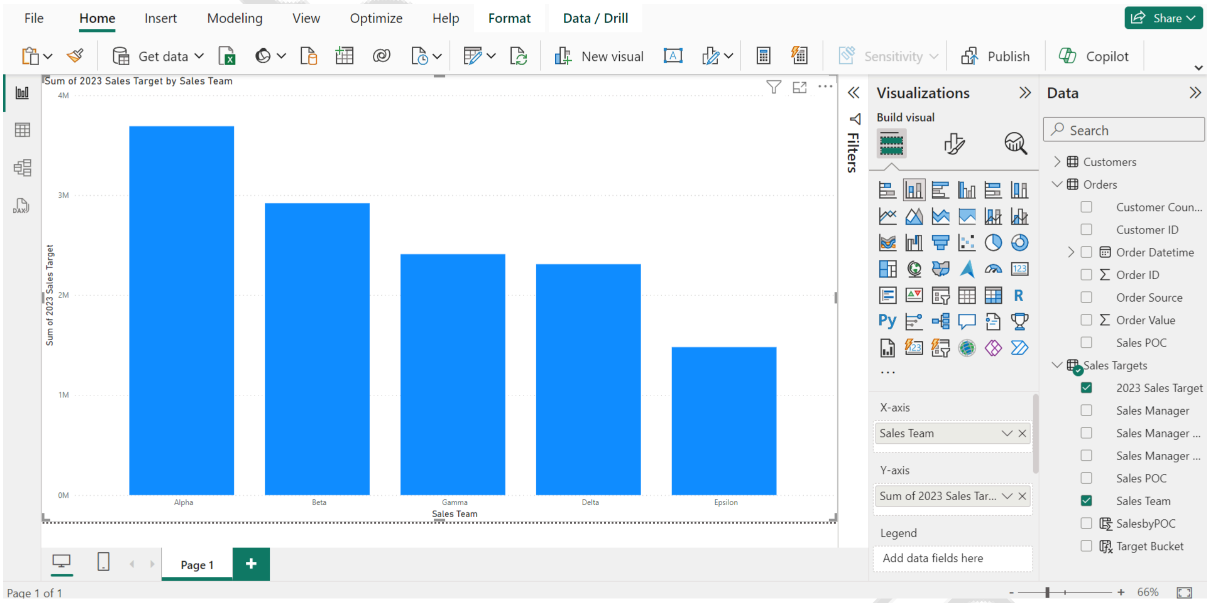
'Sales Targets'[SalesbyPOC] > 'Sales Targets'[2023 Sales Target], "Target Exceeded")



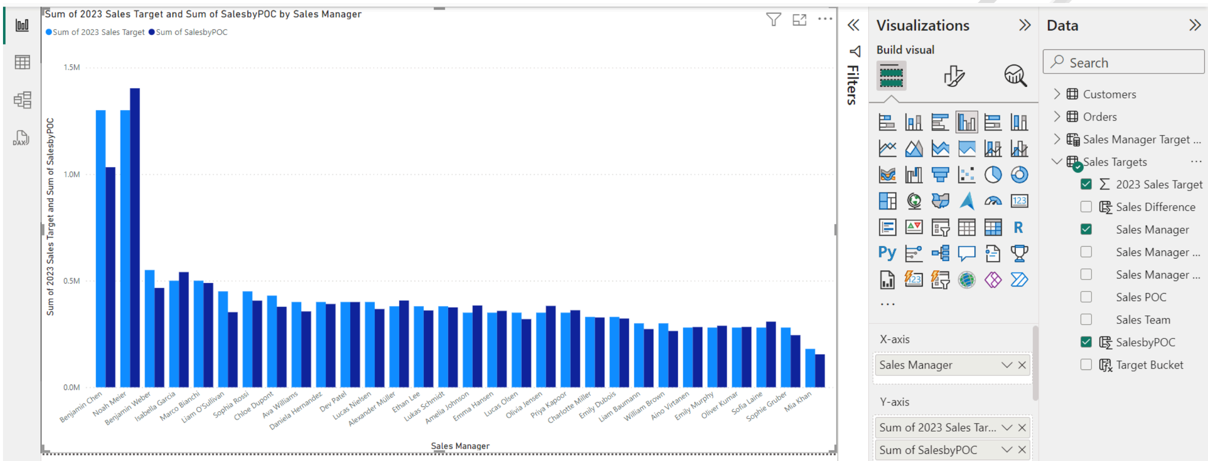
Create a chart to show these categories:



Create a chart which has bars/columns for each Sales Team:

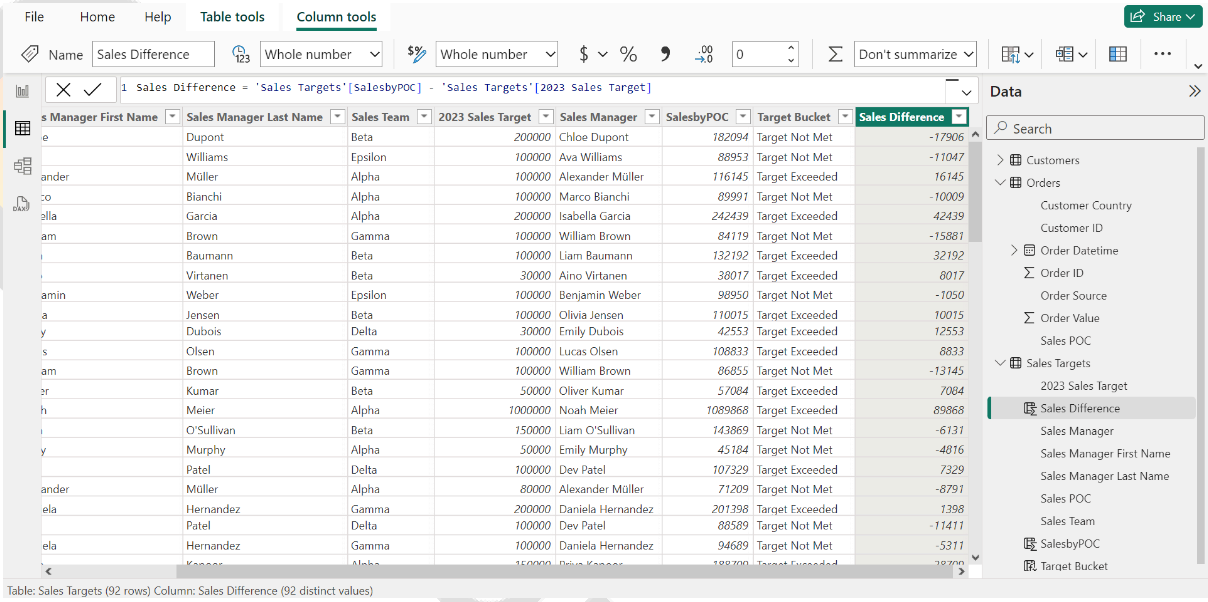


Create a chart for total target for each Sales Manager:

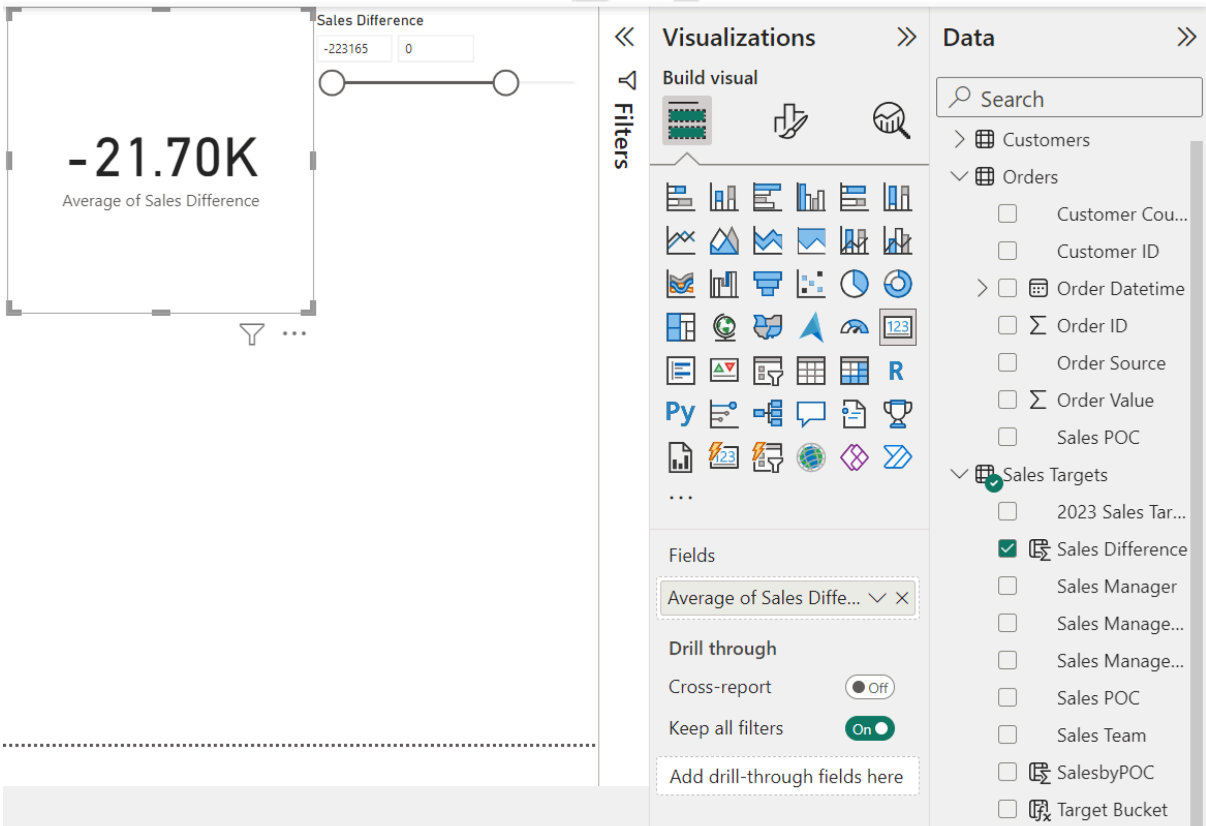


Benjamin Chen and Noah Meier were given the highest target. Benjamin Chen did not meet the sales target while Noah Meier met it.

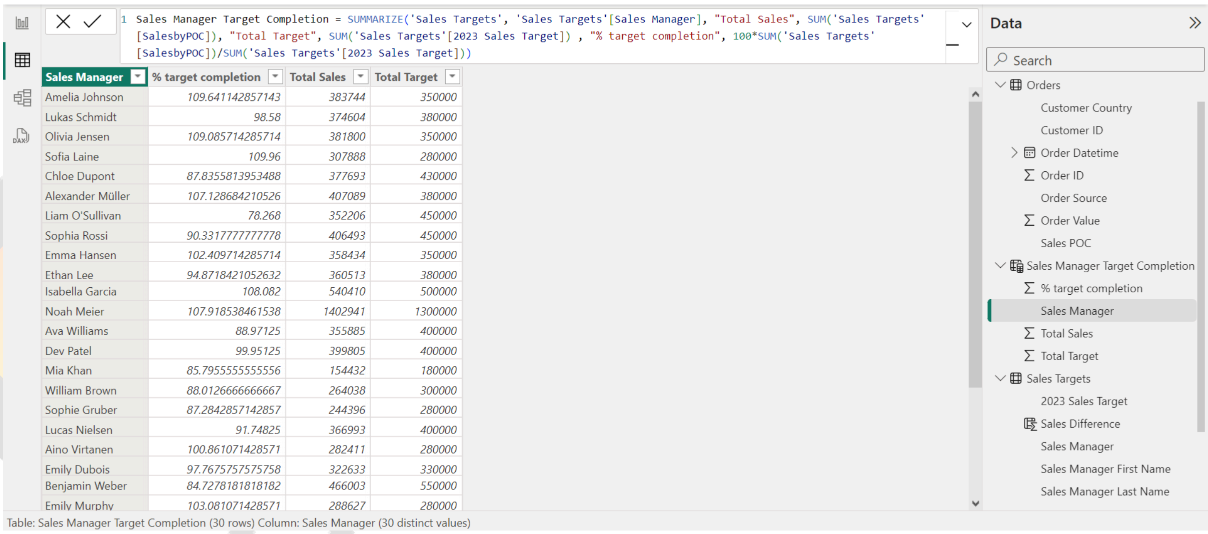
- Create a new DAX column for finding the difference between the Actual Sales and the Target. If it is negative, it means that target has not been met.



Now we need to simply filter this column for negative values and find the average. We will do this using a slicer and a Card visual for the average.



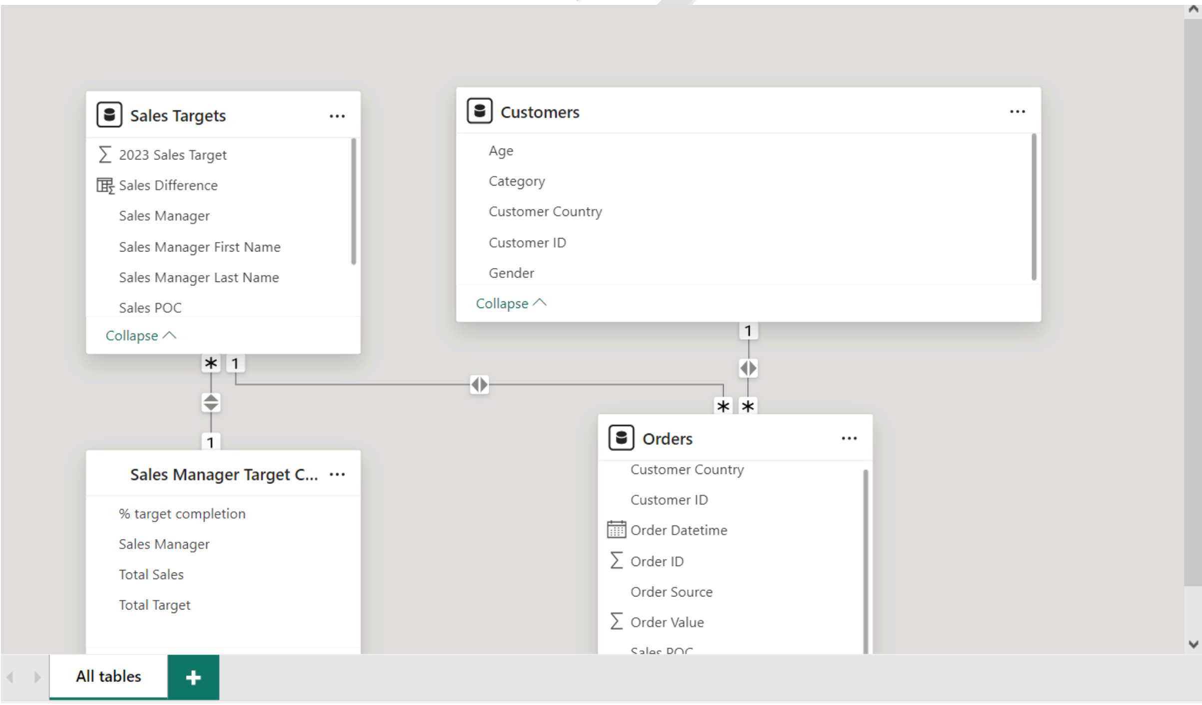
Create a new Sales Manager Targets table as shown below



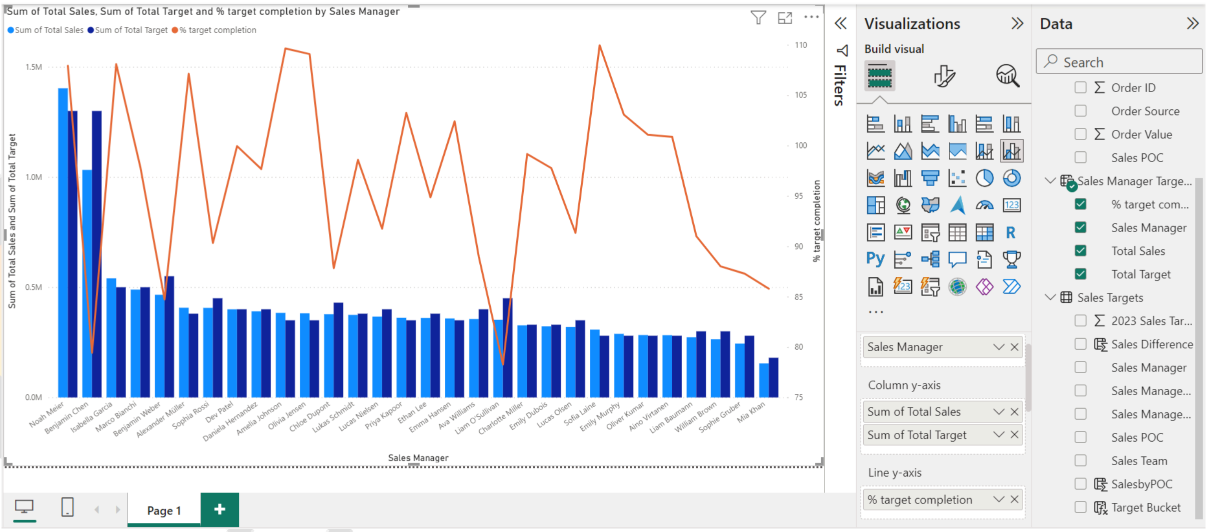
Sales Manager	% target completion	Total Sales	Total Target
Amelia Johnson	109.641142857143	383744	350000
Lukas Schmidt	98.58	374604	380000
Olivia Jensen	109.085714285714	381800	350000
Sofia Laine	109.96	307888	280000
Chloe Dupont	87.8355813953488	377693	430000
Alexander Müller	107.128684210526	407089	380000
Liam O'Sullivan	78.268	352206	450000
Sophia Rossi	90.3317777777778	406493	450000
Emma Hansen	102.409714285714	358434	350000
Ethan Lee	94.8718421052632	360513	380000
Isabella Garcia	108.082	540410	500000
Noah Meier	107.918538461538	1402941	1300000
Ava Williams	88.97125	355885	400000
Dev Patel	99.95125	399805	400000
Mia Khan	85.7955555555556	154432	180000
William Brown	88.0126666666667	264038	300000
Sophie Gruber	87.2842857142857	244396	280000
Lucas Nielsen	91.74825	366993	400000
Aino Virtanen	100.861071428571	282411	280000
Emily Dubois	97.7675757575758	322633	330000
Benjamin Weber	84.7278181818182	466003	550000
Emily Murphy	103.081071428571	288627	280000

Table: Sales Manager Target Completion (30 rows) Column: Sales Manager (30 distinct values)

Add Sales Manager Target Completion Table to the Data Model so that filters work properly. A relationship with Sales Targets table can be created (1: many, bidirectional). This is shown below.



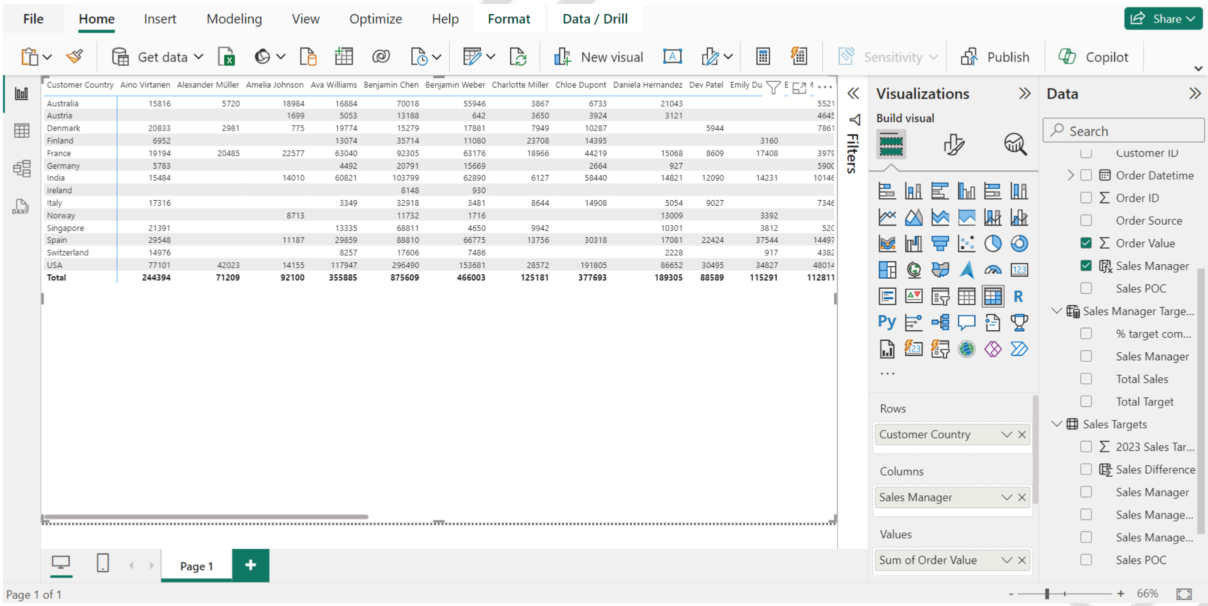
Plot the % on a chart.



From the chart above, we can see that Noah Meier had the highest Sale, and was able to exceed the target by approx ~8% as the actual sales was ~108% of the target.

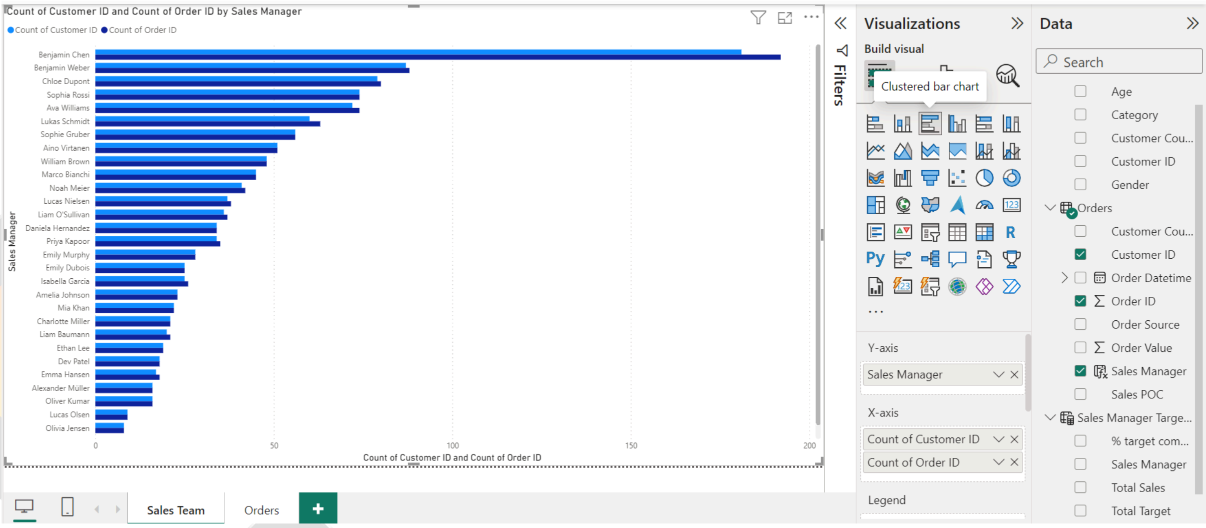
6. Let's add Sales Manager to the Orders table itself for easier creation of charts. This can be done using the RELATED function.

The matrix below can be used to answer the first part of question 6.

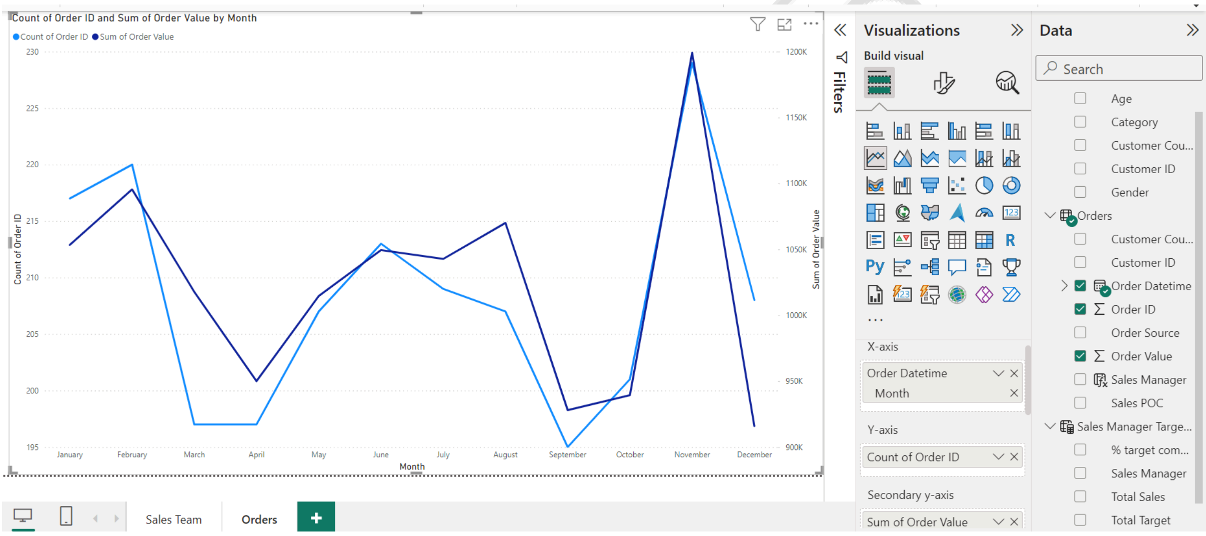


Customer Country	Aino Virtanen	Alexander Müller	Amelia Johnson	Ava Williams	Benjamin Chen	Benjamin Weber	Charlotte Miller	Chloe Dupont	Daniela Hernandez	Dev Patel	Emily Du	
Australia	15816	5720	18984	16884	70018	55946	3867	6733	21043			5521
Austria			1699	5053	13188	642	3650	3924	3121			4645
Denmark	20833	2981	775	19774	15279	17881	7949	10287		5944		7861
Finland	6952			13074	35714	11080	23708	14395			3160	
France	19194	20485	22577	63040	92305	63176	18966	44219	15068	8609	17408	3979
Germany	5783			4492	20791	15669		2664	927			5900
India	15484		14010	60821	103799	62890	6127	58440	14821	12090	14231	10146
Ireland					8148	930						
Italy	17316			3349	32918	3481	8644	14908	5054	9027		7346
Norway			8713		11732	1716			13009		3392	
Singapore	21391			13335	68811	4650	9942		10301		3812	520
Spain	29548		11187	29859	88810	66775	13756	30318	17081	22424	37544	14497
Switzerland	14976			8257	17606	7486			2228		917	4382
USA	77101	42023	14155	117947	296490	153681	28572	191805	86652	30495	34827	48014
Total	**244394**	**71209**	**92100**	**355885**	**875609**	**466003**	**125181**	**377693**	**189305**	**88589**	**115291**	**112811**

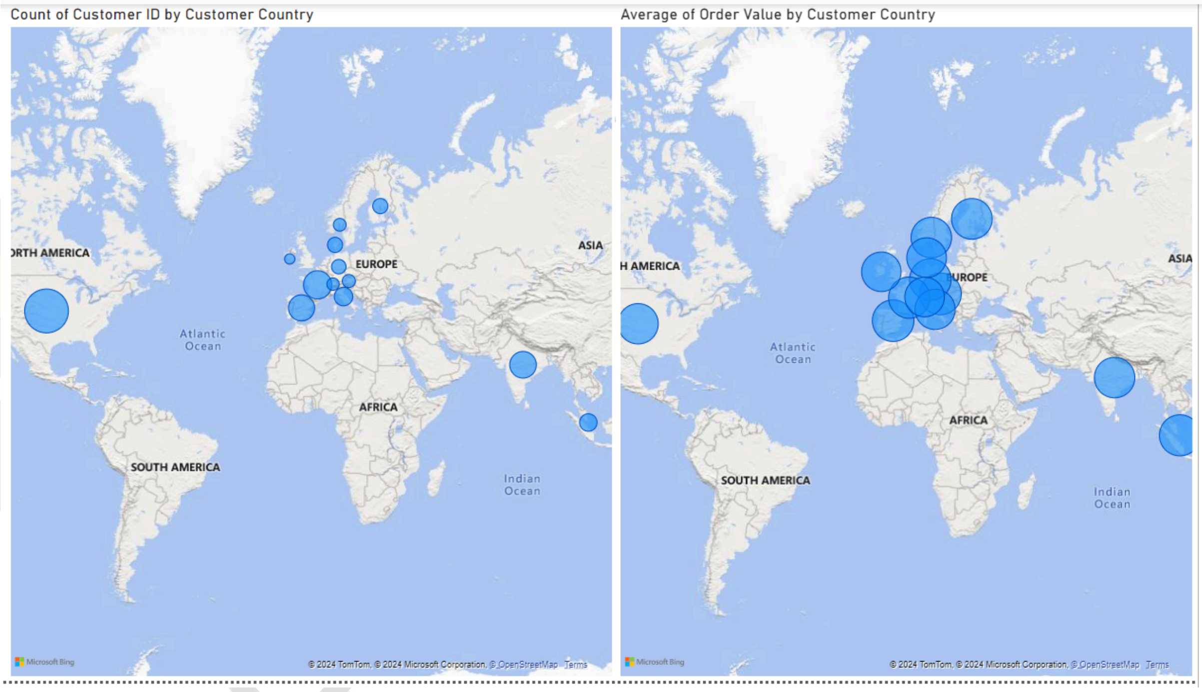
Create a chart to plot distinct count of Customer ID and order ID by Sales Manager



7. Create a line chart for demonstrating this trend



8. Plot the number of orders and average order value using map visualisations.



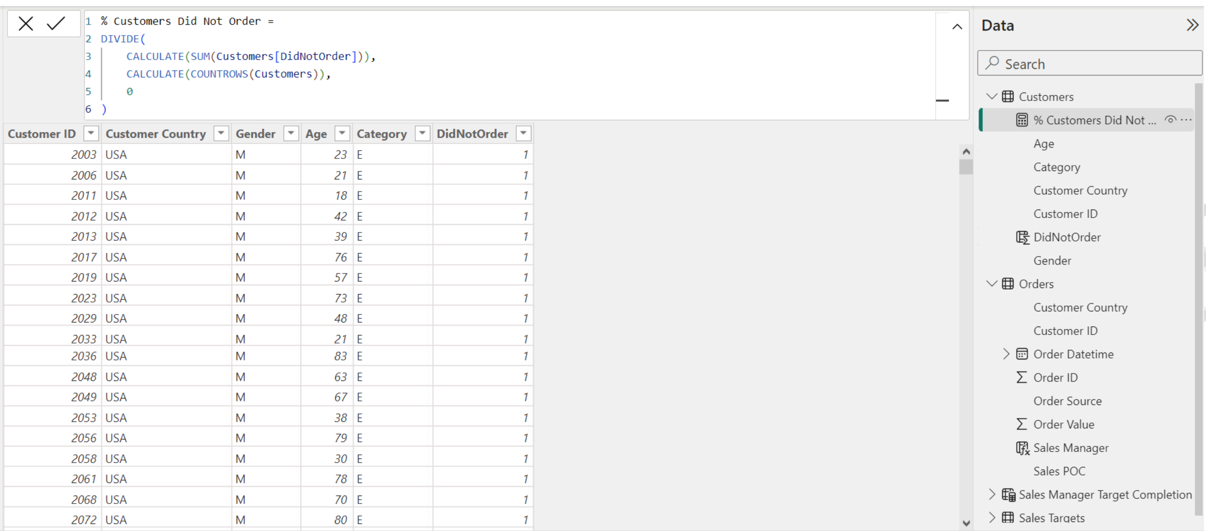
9. This one is slightly tricky and requires you to create a measure.
   a. First, create a calculated column: DidNotOrder which is 1 if there is an order corresponding to a customer ID, else 0.
   DidNotOrder = IF(COUNTROWS(RELATEDTABLE(Orders)) = 0, 1, 0)
   b. Then create a measure to calculate the % of customers who did not order.
   % Customers Did Not Order =
   DIVIDE(

   CALCULATE(SUM(Customers[DidNotOrder])),

   CALCULATE(COUNTROWS(Customers)),

   0

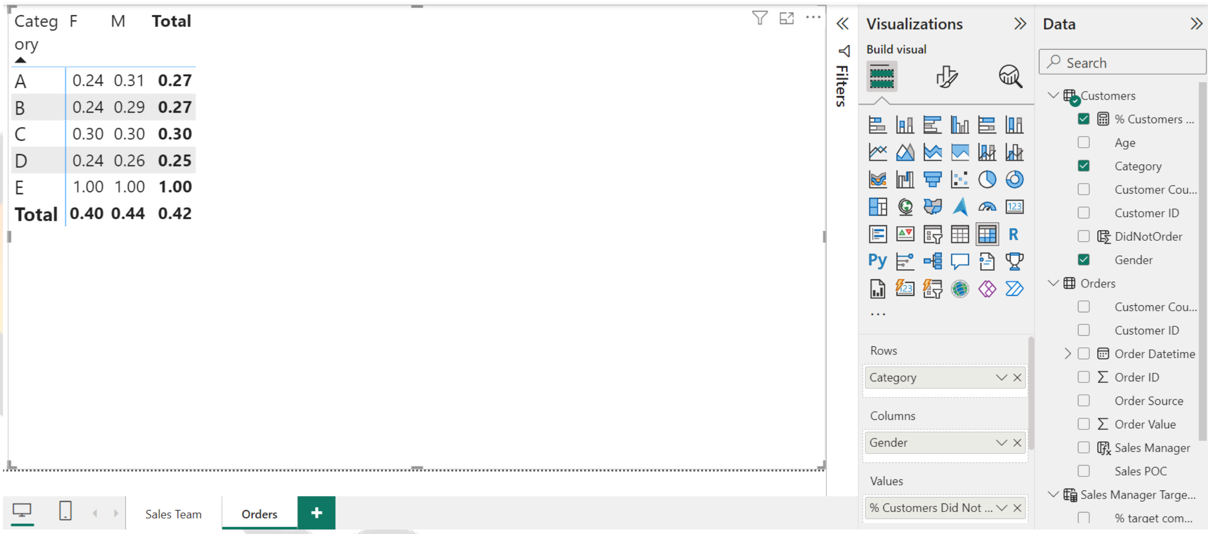
)



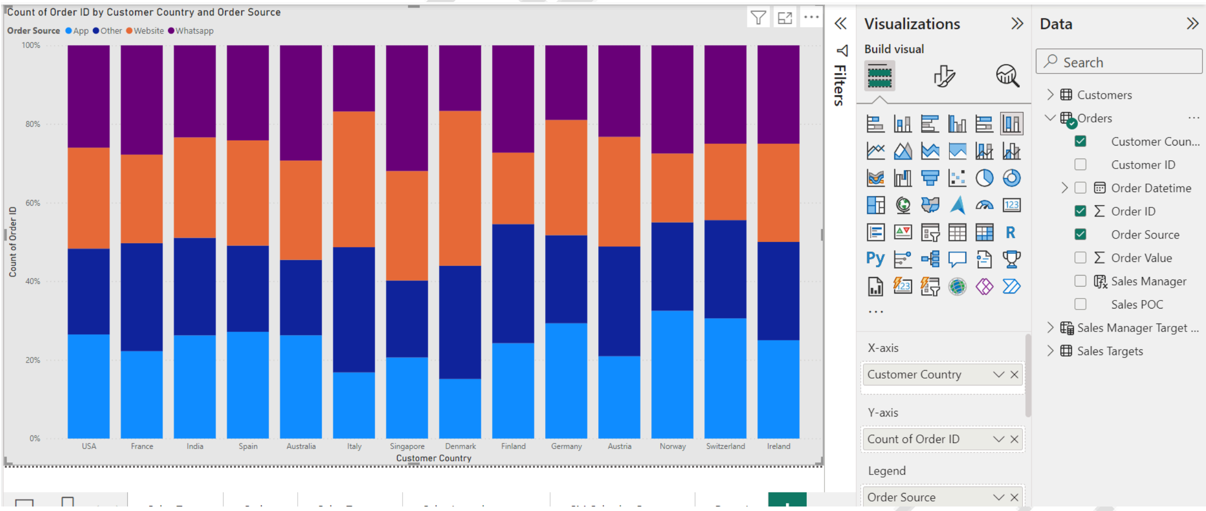
```
% Customers Did Not Order =
DIVIDE(
    CALCULATE(SUM(Customers[DidNotOrder])),
    CALCULATE(COUNTROWS(Customers)),
    0
)
```

Customer ID	Customer Country	Gender	Age	Category	DidNotOrder
2003	USA	M	23	E	1
2006	USA	M	21	E	1
2011	USA	M	18	E	1
2012	USA	M	42	E	1
2013	USA	M	39	E	1
2017	USA	M	76	E	1
2019	USA	M	57	E	1
2023	USA	M	73	E	1
2029	USA	M	48	E	1
2033	USA	M	21	E	1
2036	USA	M	83	E	1
2048	USA	M	63	E	1
2049	USA	M	67	E	1
2053	USA	M	38	E	1
2056	USA	M	79	E	1
2058	USA	M	30	E	1
2061	USA	M	78	E	1
2068	USA	M	70	E	1
2072	USA	M	80	E	1

Then create a matrix as shown below to calculate this % for each combination of gender and category.

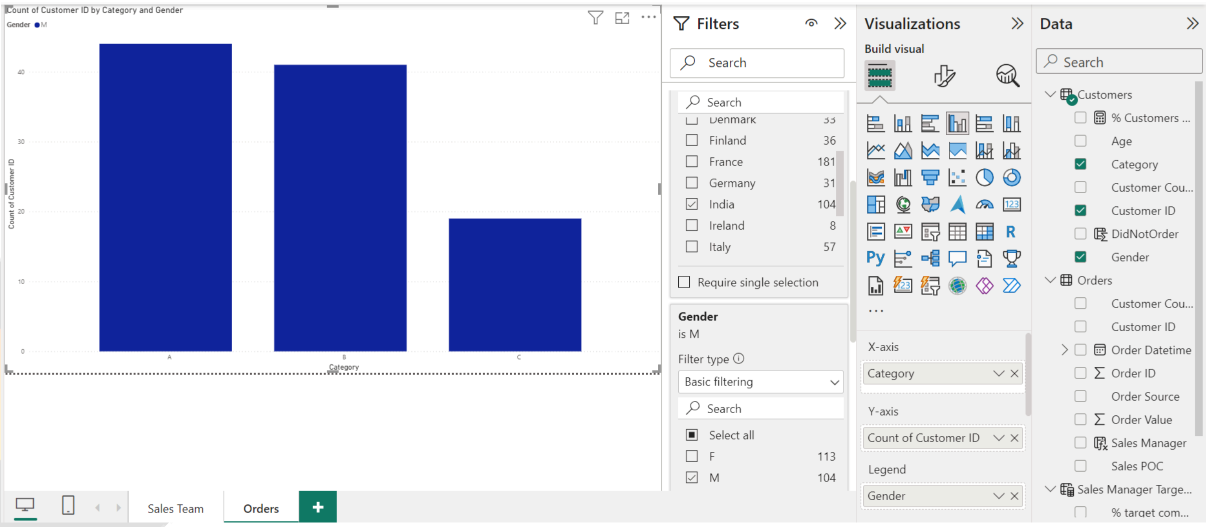


10. Plot number of orders in each country with legend as Order Source.

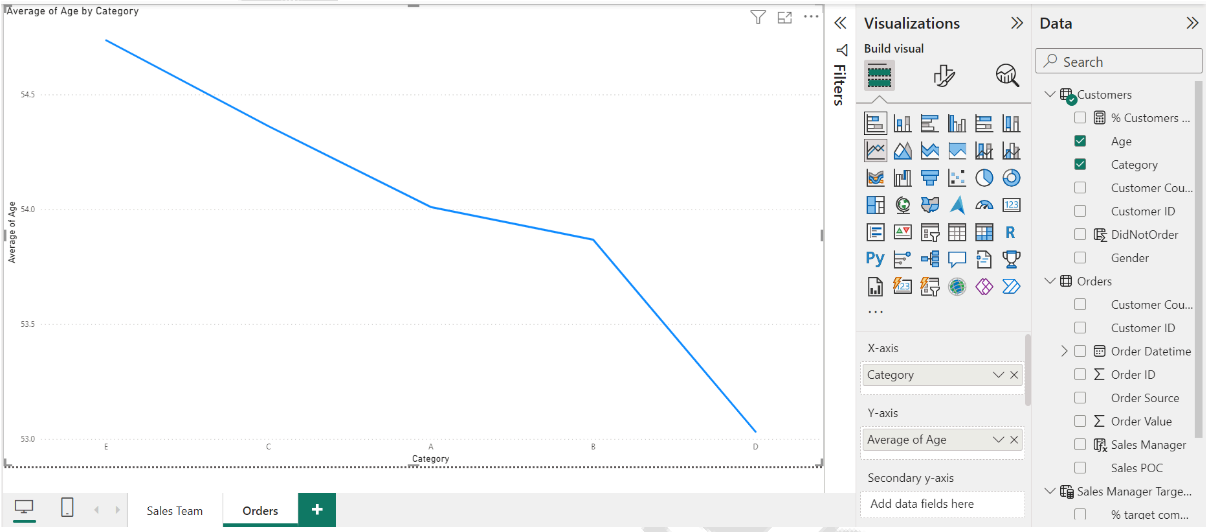


11.

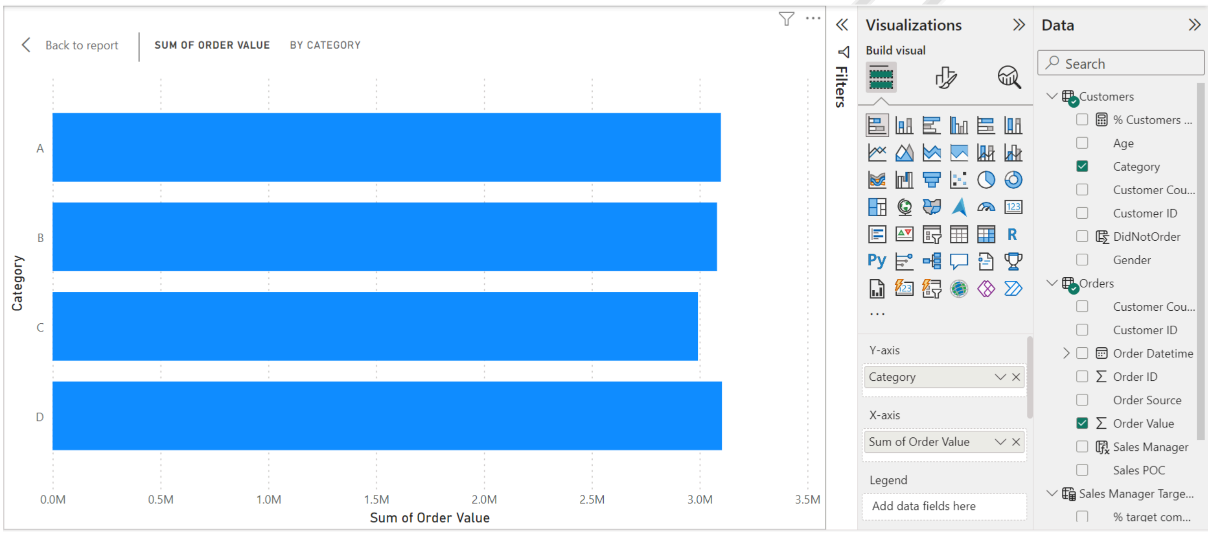
a. Plot count of customer ID by category with appropriate filters



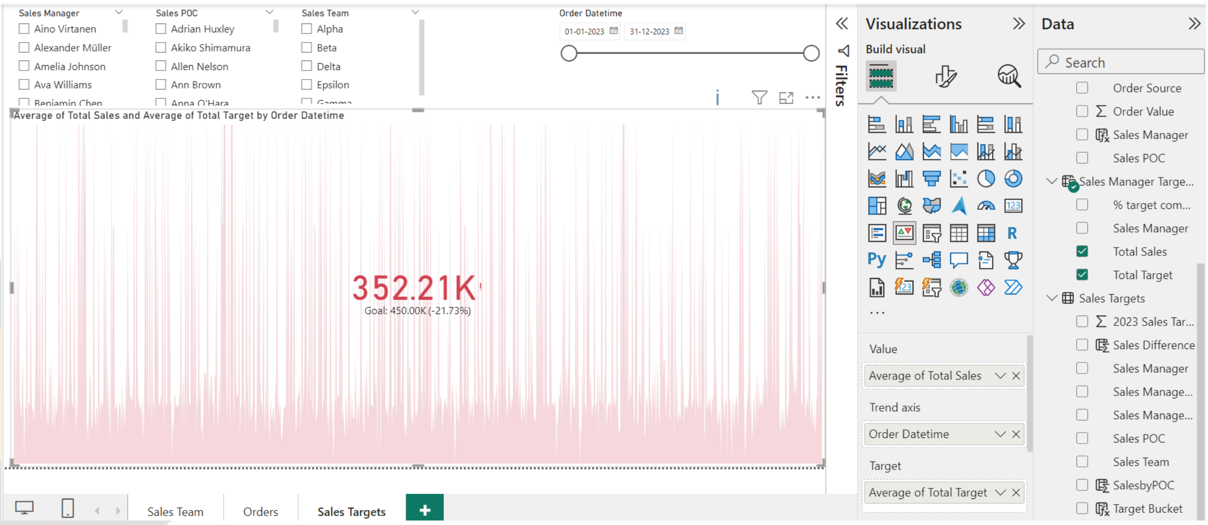
b. Find average age per category of customers:



c. Create a total order value by category chart and sort the category in Ascending order.



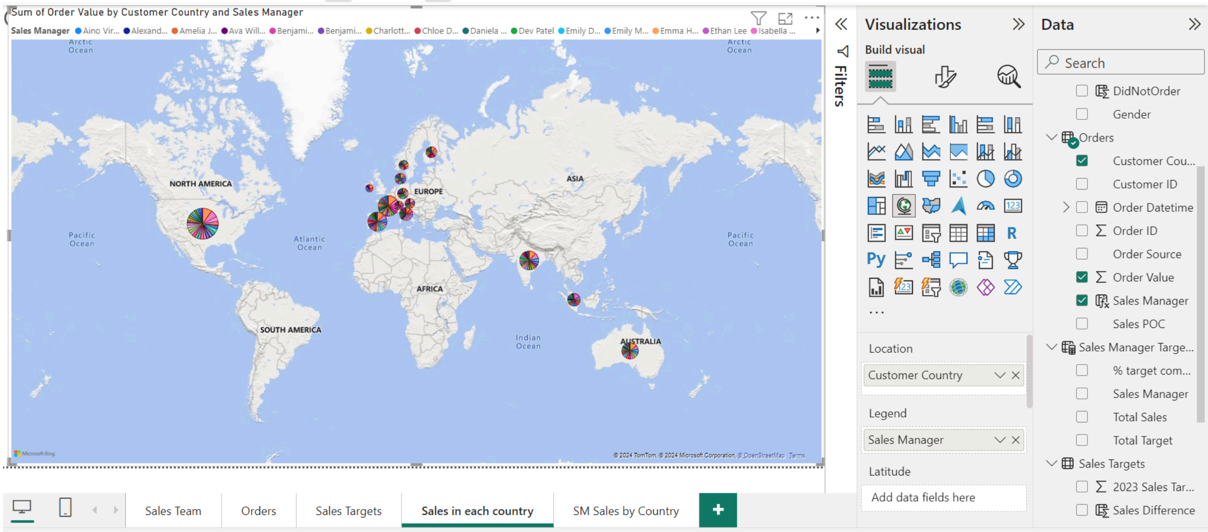
d.



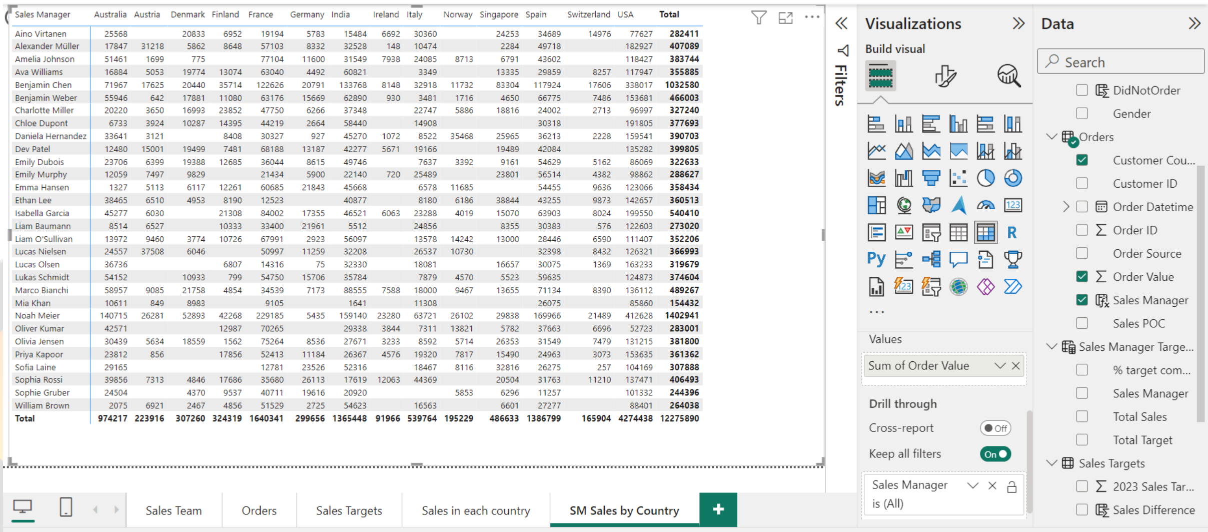
12. Add a Q&A Visual and customize the questions if required.

13.

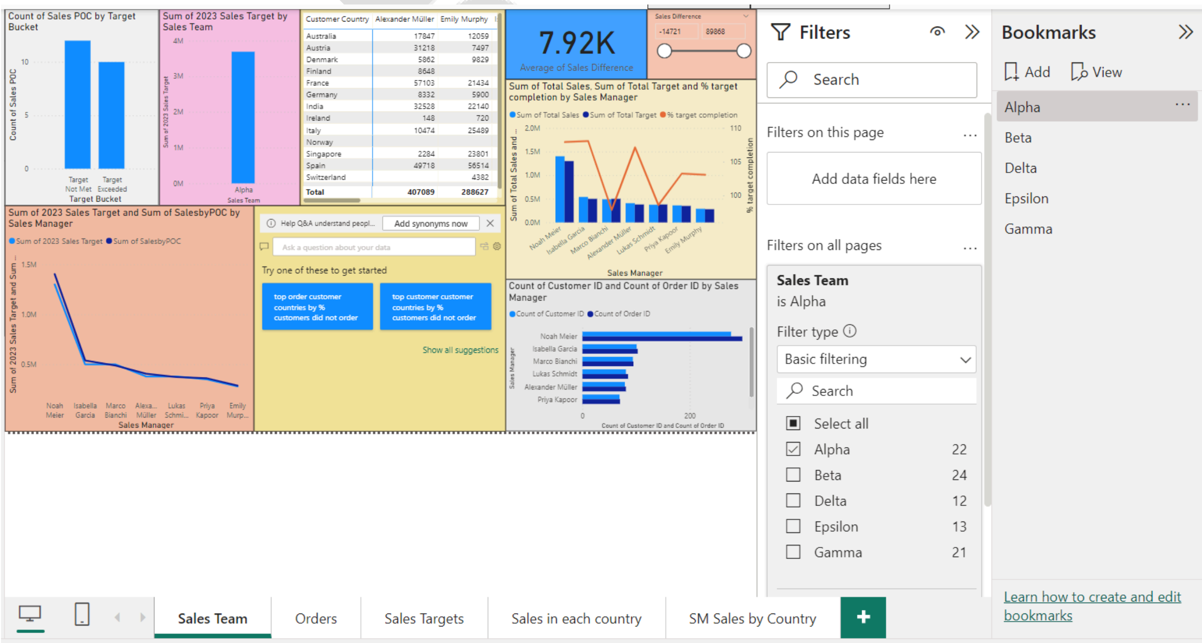
   a. Create a page with a map for total order value by country. The legend here must be the Sales Manager.



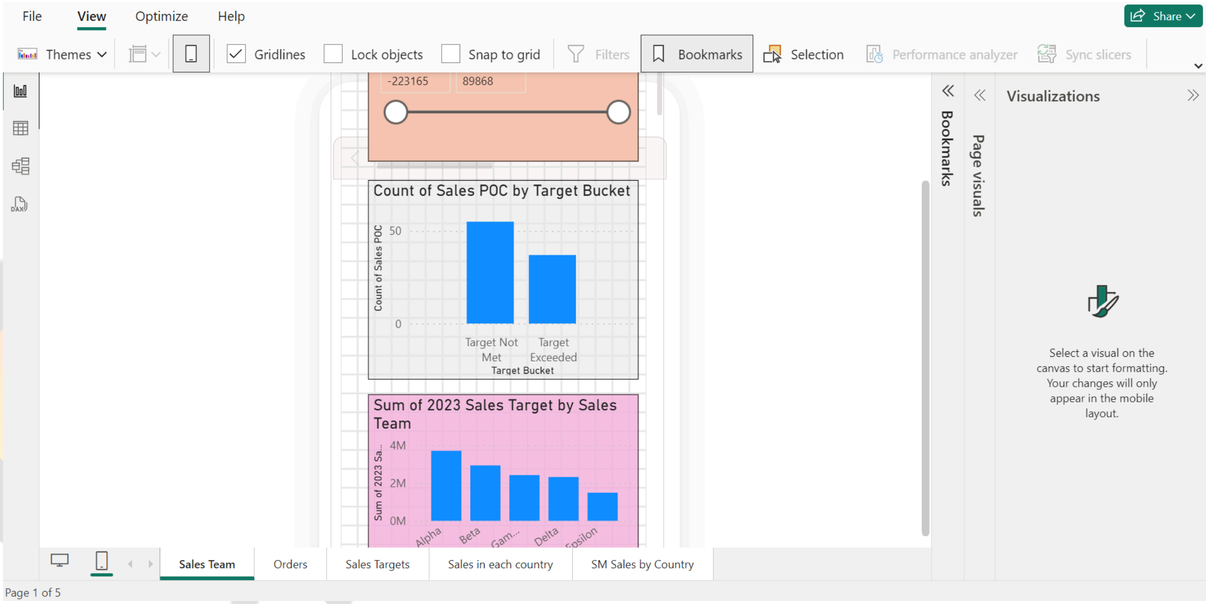
   b. Create another page for the drillthrough. On this page create a Matrix for total sales for each SM in each country. In the drillthrough fields, the Sales Manager must be added.



14. Check Solution PBIX file
15. Use Filters on all pages option to create 5 different views: one for each team and create a corresponding bookmark for each view



16. Mobile View can be created using the mobile icon at the bottom. Remember to keep visuals large and less in number on each page in the mobile view


File
View
Optimize
Help
Share
Themes
Gridlines
Lock objects
Snap to grid
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
-223165
89868
Count of Sales POC by Target Bucket
Count of Sales POC
50
0
Target Not Met
Target Exceeded
Target Bucket
Sum of 2023 Sales Target by Sales Team
Sum of 2023 Sa...
4M
2M
0M
Alpha
Beta
Gam...
Delta
Epsilon
Bookmarks
Page visuals
Visualizations
Select a visual on the canvas to start formatting. Your changes will only appear in the mobile layout.
Sales Team
Orders
Sales Targets
Sales in each country
SM Sales by Country
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